

Perform Memory Capture and Analysis Using FTK Imager

1. AIM

To capture the RAM (memory) of a Windows computer using FTK Imager and perform basic examination of the captured memory evidence.

2. SOFTWARE REQUIRED

- Windows 10/11 computer
- FTK Imager
- Sufficient free disk space
- Administrator permission on the lab computer

3. SHORT THEORY

RAM is temporary memory used by a computer while programs are running. It may contain useful temporary information such as running processes, loaded programs and other system information. This information can be lost when the computer is switched off. Therefore, memory capture is useful in digital forensics.

4. IMPORTANT NOTE

Perform this experiment only on a college laboratory computer, your own computer, or a computer for which you have permission. Memory capture should not be performed on someone else's computer without authorization.

5. INSTALL FTK IMAGER

FTK Imager is a forensic imaging and evidence examination tool from Exterro / AccessData.

1. Open a web browser such as Chrome or Edge.
2. Search for the official FTK Imager download page.
3. Open the official Exterro FTK Imager download page.
4. Download the Windows installer provided for FTK Imager.
5. After downloading, open the installer file.
6. If Windows shows a User Account Control message, select Yes if you are authorized to install software on the lab computer.
7. Follow the installation wizard.
8. Accept the license agreement when required.
9. Keep the default installation location unless your instructor specifies another location.
10. Click Install and wait until installation is completed.
11. Click Finish.

Note: The exact download page and installer screen can change between FTK Imager releases. Always use the official vendor source.

6. HOW TO OPEN / RUN FTK IMAGER

1. Click the Windows Start button.
2. Type FTK Imager.
3. Select FTK Imager from the search results.
4. If your lab requires administrator privileges, right-click FTK Imager and choose Run as administrator.
5. Wait for the FTK Imager main window to open.
6. You should see the FTK Imager interface with menus and evidence/source panels.

7. PREPARE FOR MEMORY CAPTURE

1. Create a folder for the practical, for example: C:\FTK_Lab\MemoryCapture
2. Make sure the destination drive has enough free space.
3. Do not switch off or restart the computer before memory capture is completed.
4. Note the computer name, Windows version and RAM size if required by your instructor.
5. Close unnecessary applications, but keep FTK Imager running.

8. MEMORY CAPTURE – STEP BY STEP

This is the main practical procedure.

1. Open FTK Imager.
2. Click the File menu.
3. Look for the option named Capture Memory or the memory-capture option provided by your version.
4. Click the memory-capture option.
5. A Memory Capture window will open.
6. Select the destination folder created for this experiment.
7. Enter a file name, for example: Memory_Capture_Student01.
8. If your version shows an option to capture the Windows pagefile, follow your instructor's instruction. If it is not required, leave it disabled.
9. Check the destination path and file name carefully.
10. Click Capture Memory / Start / Capture, according to the wording in your version.
11. Wait until the capture finishes.
12. Do not shut down or restart the computer during the capture.
13. When the capture is complete, note the completion message.
14. Close the capture window.

9. CHECK THE CAPTURED FILE

1. Open File Explorer.
2. Go to the folder where you saved the memory capture.
3. Check that the captured memory file is present.
4. Check the file name and file size.
5. Do not delete or edit the captured file.
6. Record the filename and file size in the observation table.

10. BASIC ANALYSIS / EXAMINATION

FTK Imager versions differ in the type of memory evidence they can directly display. If your version supports examination of the captured memory source, use the following basic procedure.

1. Open FTK Imager.
2. Use the option for adding an evidence source.
3. Select the appropriate source type for the captured memory file, if available.
4. Browse to the memory-capture file.
5. Add the evidence source.
6. Expand the evidence tree.
7. Observe the information displayed by FTK Imager.
8. Record the available information in the observation table.
9. If your FTK Imager version does not directly analyse the RAM image, keep the captured image safely and use a dedicated memory-analysis tool such as Volatility for the analysis stage, if your syllabus requires it.

11. WHAT TO OBSERVE

- Memory capture file name
- Memory capture file size
- Date and time of capture
- Computer name
- Windows version
- RAM size
- Any information displayed by FTK Imager
- Any error or warning message

12. OBSERVATION TABLE

Parameter	Observation
FTK Imager Version	
Windows Version	
Computer Name	
RAM Size	
Capture Date	
Capture Time	
Memory File Name	
Memory File Size	
Analysis / Findings	
Remarks	

13. RESULT

Thus, the memory (RAM) of the authorized Windows computer was successfully captured using FTK Imager and the captured evidence was examined for the information available in the installed FTK Imager version.

14. PRECAUTIONS

- Use only an authorized computer.
- Run the tool with the required permissions.
- Make sure enough storage space is available.
- Do not restart or shut down the computer during memory capture.
- Do not modify or delete the captured memory file.
- Record the filename, size and capture time.
- Use the official FTK Imager download source.

15. COMMON PROBLEMS

Problem	Solution
FTK Imager does not open	Try Run as administrator or ask the lab administrator to provide the required permission.
Memory capture option is not visible	Check the installed FTK Imager version and use the memory-capture feature available in that version.
Capture fails	Check free disk space, permissions and the destination path.
Captured file is not visible	Check the destination folder and confirm that the capture completed successfully.
Cannot analyse the memory image in FTK Imager	This can be a tool/version limitation. Preserve the image and use a dedicated memory-analysis tool if required by the practical.

16. VIVA QUESTIONS

Q. What is RAM?

A. RAM is temporary memory used by a computer to store information required by running programs and the operating system.

Q. What is memory capture?

A. It is the process of acquiring the contents of volatile memory into an evidence file.

Q. Why is RAM important in forensics?

A. RAM can contain temporary information that may disappear after shutdown.

Q. What is FTK Imager?

A. FTK Imager is a digital-forensics tool used for evidence acquisition and examination.

Q. Why should we not shut down the computer before memory capture?

A. Shutting down can cause volatile information in RAM to be lost.

Q. Why do we need free disk space?

A. The captured memory image can be large and needs storage space.

Q. Can the memory capture process change RAM?

A. Yes. Running any capture program uses some memory, so live acquisition can affect the system slightly.

Q. What is the purpose of recording the capture time?

A. It helps document when the memory snapshot was taken.

Q. What should be done with the original memory image?

A. It should be preserved and not unnecessarily modified.

Q. Which tool can be used for detailed RAM analysis?

A. A dedicated memory-forensics tool such as Volatility can be used.

17. SCREENSHOTS TO TAKE FOR RECORD

1. FTK Imager installed and opened
2. FTK Imager main window
3. Memory Capture option/window
4. Memory capture destination and filename
5. Memory capture completed message
6. Captured memory file in the destination folder
7. Evidence/analysis screen, if supported